

Main energy levels and sublevels

Learning outcomes

By the end of this section you should be able to:

- (S1.3.3) Deduce the maximum number of electrons that can occupy each energy level.
- (S1.3.4) Recognise the shape and orientation of an s atomic orbital and the three p atomic orbitals.

In [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) you learned that electrons are located in discrete energy levels called main energy levels, or principal energy levels. A rather interesting thing about electrons is that, because of their very small size and mass, we cannot know exactly where they are located. In 1927, a German physicist named [Werner Heisenberg](https://en.wikipedia.org/wiki/Werner_Heisenberg) (https://en.wikipedia.org/wiki/Werner_Heisenberg) came up with the [Heisenberg uncertainty principle](#) which, in its simplest form, states that it is not possible to know both the location and [velocity](#) of very small particles such as electrons. So how do we overcome this? Rather than knowing exactly where an electron is, we assign regions of space, known as atomic orbitals, where we can be fairly certain that electrons are located. The Heisenberg uncertainty principle is outside of the scope of the IB chemistry syllabus and is not assessed, but if you are interested, you may read about it in more detail [here](#)  (<https://www.theguardian.com/science/2013/nov/10/what-is-heisenbergs-uncertainty-principle>).

Making connections

In [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) you learned that the electrons are located in energy levels within the atom labelled $n = 1$, $n = 2$, etc. If you need to review this section then now might be a good time to do so.

Main energy levels

The main energy levels were introduced in [section S.1.3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>). Recall that each energy level is assigned a number, starting with $n = 1$ with the lowest energy and being closest to the nucleus. As the value of n increases, the energy increases and so does the distance from the nucleus. How do you think the electrons are organised in these energy levels? Do you think they are randomly arranged or is there a system of organisation?

To review the concept of main energy levels, try **Interactive 1** before you continue with this section. In the first part, drag the labels into the correct boxes. In the second part, fill in the blanks to complete the sentences.

Diagram illustrating the main energy levels of an atom. The nucleus is represented by a green dot in the center. Concentric circles represent the energy levels. Labels for the energy levels ($n = 1$, $n = 2$, $n = 3$, $n = 4$, $n = 5$) and the Nucleus are provided in boxes to the right of the diagram. A 'Check' button is located at the bottom left, and a right arrow is at the bottom right.

Interactive 1. Energy Levels Drag and Drop.

The main energy levels are divided into sublevels. The sublevels are assigned the letters s, p, d and f based on the shape of atomic orbitals they contain. Atomic orbitals can be thought of as regions of space where there is a high probability of finding an electron. A single atomic orbital can hold a maximum of two electrons and as you will learn in section S1.3.5

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>), they must have specific orientations. So, to recap, we have main energy levels which are divided into sublevels which are made up of atomic orbitals (**Figure 1**).

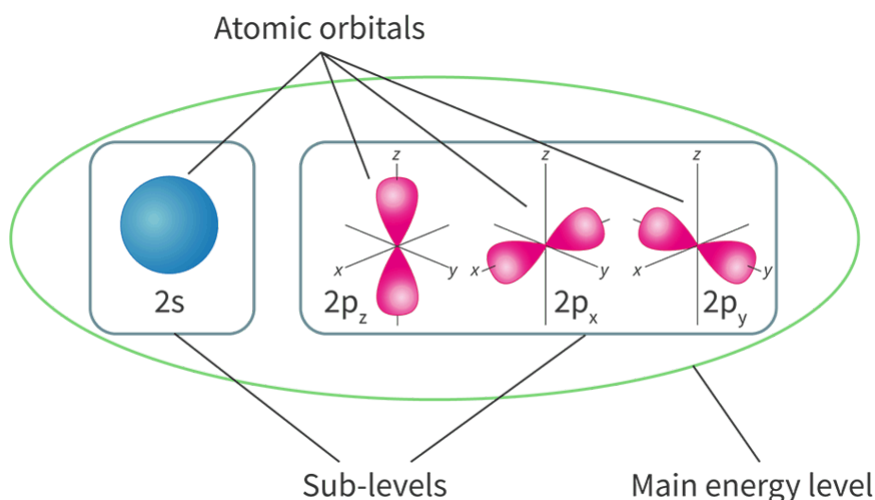


Figure 1. The shapes of s and p atomic orbitals.

 More information for figure 1

The image shows a diagram illustrating the shapes of s and p atomic orbitals. On the left, the s orbital is represented as a blue sphere indicating its spherical shape. On the right, the p orbitals are depicted in pink with a dumbbell shape, demonstrating their characteristic form. The p orbitals are shown in three orientations, commonly labeled as p_x , p_y , and p_z , reflecting their spatial orientation along the x, y, and z axes. This image explains the correlation between these orbital shapes and their electron-holding capacities: the s orbital, being singular, holds up to two electrons, whereas the three p orbitals collectively hold up to six electrons. The surrounding text provides additional context on energy levels and sublevels.

[Generated by AI]

Figure 1 shows the shapes of two specific atomic orbitals: s atomic orbitals and p atomic orbitals. The s atomic orbitals are spherical and p atomic orbitals are dumbbell shaped. An s sublevel only consists of a single s atomic orbital so it can hold a maximum of two electrons. A p sublevel is composed of three p atomic orbitals so it can hold a maximum of six electrons.

There are also d and f atomic orbitals but the shapes of these are complex and are not covered in the IB chemistry course. A d sublevel contains five d atomic orbitals, and an f sublevel contains seven f atomic orbitals. Regardless of the type of atomic orbital (s, p, d or f) a single orbital can hold a maximum of two electrons. **Table 1** shows the first four main energy levels, together with the number of sublevels, atomic orbitals and total number of electrons.

Table 1. The first four main energy levels and orbitals

Main energy level, n	Sublevels	Number of s orbitals	Number of p orbitals	Number of d orbitals	Number of f orbitals
1	1s	1	0	0	0
2	2s, 2p	1	3	0	0
3	3s, 3p, 3d	1	3	5	0
4	4s, 4p, 4d, 4f	1	3	5	7

From the table, you can see that the maximum number of electrons in an energy level is $2n^2$ where n is the principal energy level number. For the first main energy level, $n = 1$, the maximum number of electrons it can hold is $2(1)^2$, which is 2. For the second main energy level, it is $2(2)^2$, which is 8. The number of electrons increases as the value of n increases. So the $n = 3$ energy level can hold a maximum of 18 electrons and the $n = 4$ energy level a maximum of 32 electrons.

Making connections

How does an element's highest main energy level relate to its period number in the periodic table? In [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/), you will learn about the relationship between the highest occupied main level of an element and its period number in the periodic table.

In this activity, you will determine the number of atomic orbitals in each of the main energy levels together with the shapes of the s and p atomic orbitals.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Reflecting at all stages of the assessment and learning cycle
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Answer the following questions.

Check your answers once you have completed all of the questions.

1. Copy and complete the table with the maximum number of electrons in each main energy level.

Principal energy level (n)	Maximum number of electrons
1	
2	
3	
4	

2. Draw a table like the one below.

Sketch the shape of an s and p atomic orbital in the table.

s orbital	

s orbital	

3. State the number of orbitals in each of the following sublevels.

- (a) The 2s sublevel
- (b) The 3p sublevel
- (c) The 3d sublevel
- (d) The 4f sublevel

4. State the total number of each type of atomic orbital (s, p, d, f) in each main energy level.

- (a) $n = 1$
- (b) $n = 2$
- (c) $n = 3$
- (d) $n = 4$

1.

Principal energy level (n)	M
1	
2	
3	
4	

2.

s orbital	p orbital
	



3. (a) The 2s sublevel (one s orbital)
(b) The 3p sublevel (three p orbitals)
(c) The 3d sublevel (five d orbitals)
(d) The 4f sublevel (seven f orbitals)
4. (a) $n = 1$ (one s orbital)
(b) $n = 2$ (one s orbital and three p orbitals)
(c) $n = 3$ (one s orbital, three p orbitals and five d orbitals)
(d) $n = 4$ (one s orbital, three p orbitals, five d orbitals and seven f orbitals)